Kiet Nguyen

CSCE 483

Notebook.

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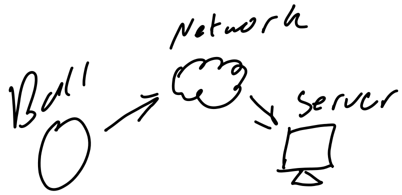
Understand how the project works.

Need to see Ball



What can be improved?

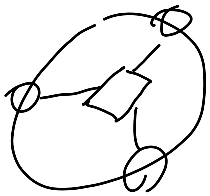
Speed/Range/Efficiency/

3 Pis

↓ prof says to reduce to 1

1

⟹ So now we need to multiplexer the cameras



Cameras are probable equidistant

Is it supposed to be rolled or thrown?

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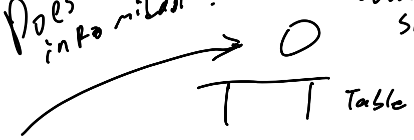
If ball is thrown

Does it gather info midair?

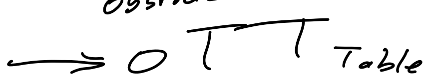
or only when it stops

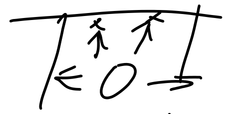


Need ball to test current state.

If ball is rolled

Isn't it easily obstructed?





Cant see

Low visibility (Smoke?)

How to see better? (Lidar?)

sit on this for now.

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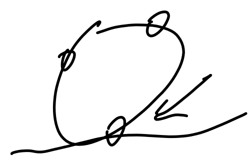
If ball rolls like this



Botton camera doesn't see anything

Cull the camera to save resources

How to determine this though?

Position Based

Always culls the gnd camera

How to tell?

Would it mess withe weight distrobution and thus rolling?

can be Horz

Not always on the ground

What about walls

Ball has

stand offs for the 3 Pis

Will need to readjust for 1

"Vision Based"

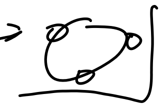
If camera sees nothing / Darkness Cull it.

How to determine? Based on light?

If all 3 cams see doen't lose sight

If more than 1 cant see, culls multiple

only 1 sees



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Consider material for heat restance (battery)

old Report says it should be fine though.

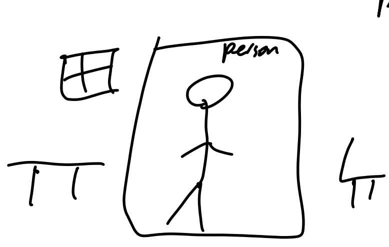
Improve slides and understanding of the project

Consider object detection?

\+

Camera Culling Alg

= potentially high load on Raspberry Pi? would need to stress test to know for sure.



Recognition software exists, but not quite sure how it recognizes people.

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Camera
Normal vs. NoIR

Same size
similar price

already have

How much better does it see? Through smoke?
Must test.
but where to get smoke?

Raspberry Pi
4B vs 5

Power diffs
$20 more exp.
2x processing power?

2/18

Get 1 NoIR camera for testing

use incense in clear box?

Raspberry Pi 4 is cheaper, more comput with battery,

3/7 — 3/14 Spring break

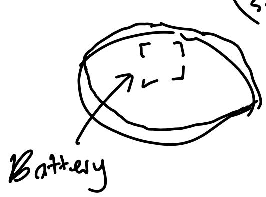
Adjust ball using AutoCAD

Cameras same
No longer need multiple pi spots,

Need spot for battery + gyro

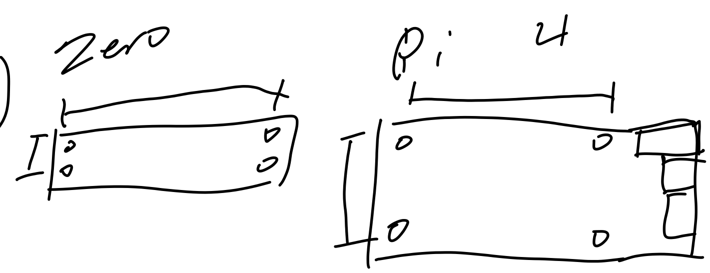
(has screw holes so standoff is ok)

Maybe corners like to hold battery?



Camera dims are same

Pi4's longer than Zero



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Thoughts on adjust ball latch mechanism

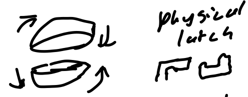
Right now

Screws down all the way

Secure, but tedious to open

Ideas

"pokeball style"



Twist to close

More convenient for us

More intuitive

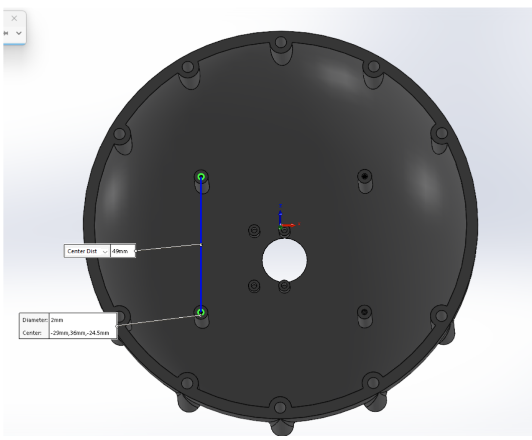
How does this affect resistance to elements?

Is this more/less secure?

Do we have effort to spare?

i.e should we even bother?

3/23 - 3/25



First ball made

Keep original ball design to fit with other half

extend for pi 4

add threads

3/30 - 4/1

Help with Mast3R-SLAM

All compiles on my end without pyTorch (cuda library)

Can't use my setup bc I don't have NVIDIA GPU → No Cuda

Packages will not download properly on brandon's setup

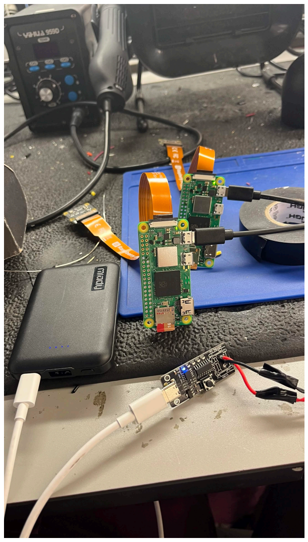
4/6 - 4/8

Adjusted ball CAD to fit gyroscope

Need to do FEDC quiz for team for printing

technical difficulties

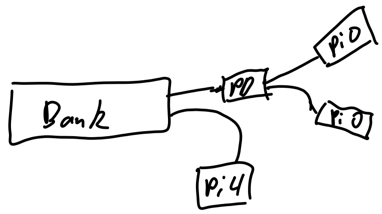
4/13 - 4/15



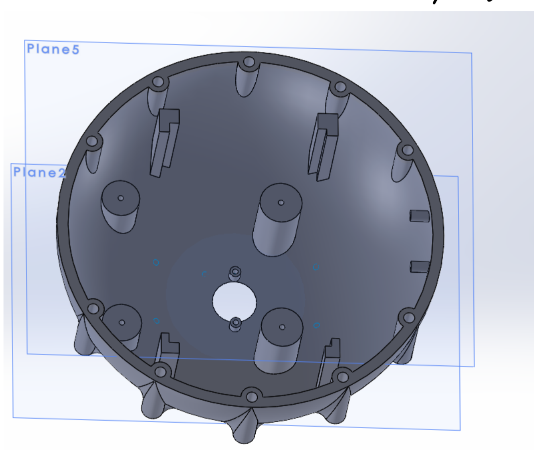
All components assembled

Reverted back to using Pi zeroes for processing power

Power distribution board allows powering both with one wire.



4/20 - 4/22



Prev Design failed drop test

Remove threading

thicken pi4 Bases

extend gyroscope stands slightly

extra standoffs for battery to sit on